



Method to derive genotypes for loci associated with the risk of Parkinson's Disease and the summarized score for the genetic burden from whole genome sequenced data

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Summary (or Abstract) [Heading 1]

This is secondary data from the July 2018 build of PPMI whole genome sequenced data. The genotypes of loci associated with the risk of Parkinson's disease were derived from that data. These loci were identified by the largest case-control study to date (Nalls et al.). The cumulative burden of the risk were calculated as genetic risk score (GRS).

Method

1. Genotypes

Genotypes of 90 loci were extracted using Plink.

2. GRS

The genetic risk scores were calculated with the default setting of "--score" option of Plink. The weights for the effect allele were given as below. (Nalls et al.)

Please note that the weights were derived from the case-control study which only included participants of European ancestry. The GRS might not reflect genetic burdens for people from other populations.

RSID	Effect Allele	Beta	Nearest Gene
rs114138760	C	0.2812	PMVK
rs35749011	A	0.6068	KRTCAP2 (high LD with GBA E326K)
rs76763715	T	-0.7467	GBAP1 (GBA N370S)
rs6658353	C	0.065	FCGR2A (high LD with FCGR2A p.His167Arg)
rs11578699	T	-0.0704	VAMP4
rs823118	T	0.1066	NUCKS1
rs11557080	A	0.1315	RAB29
rs4653767	T	0.0833	ITPKB
rs10797576	T	0.1114	SIPA1L2
rs76116224	A	0.1104	KCNS3
rs2042477	A	-0.0657	KCNIP3





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rs11683001	A	0.0705	MAP4K4
rs57891859	A	0.0807	TMEM163
rs1474055	T	0.1796	STK39
rs73038319	A	-0.1693	SATB1
rs6808178	T	0.0658	LINC00693
rs12497850	T	0.0636	IP6K2
rs55961674	T	0.0861	KPNA1
rs11707416	A	-0.0627	MED12L
rs1450522	A	-0.0616	SPTSSB
rs10513789	T	0.1485	MCCC1
rs873786	T	-0.1731	GAK (high LD with TMEM175 p.Q65P)
rs34311866	T	-0.2126	TMEM175 (TMEM175 p.M393T)
rs4698412	A	0.1035	BST1
rs34025766	A	-0.0839	LCORL
rs6825004	C	0.0622	SCARB2
rs4101061	A	-0.0912	FAM47E
rs6854006	T	-0.0912	FAM47E-STBD1
rs356182	A	-0.2774	SNCA
rs5019538	A	-0.1565	SNCA
rs13117519	T	0.0875	CAMK2D
rs62333164	A	-0.0638	CLCN3
rs1867598	A	-0.1554	ELOVL7
rs26431	C	0.0621	PAM
rs11950533	A	-0.0916	C5orf24
rs4140646	A	0.0833	LOC100131289
rs9261484	T	-0.0635	TRIM40
rs112485576	A	-0.1676	HLA-DRB5
rs12528068	T	0.0657	RIMS1
rs997368	A	0.0714	FYN
rs75859381	T	-0.2207	RPS12 (high LD with SLC18B1 p.S30P)
rs199351	A	0.1016	GPNMB
rs76949143	A	-0.1432	GS1-124K5.11
rs1293298	A	0.093	CTSB
rs620513	T	-0.0856	FGF20
rs2280104	T	0.0556	BIN3
rs2086641	T	-0.0605	FAM49B
rs13294100	T	-0.0859	SH3GL2
rs10756907	A	-0.0926	SH3GL2
rs6476434	T	-0.0615	UBAP2

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rs896435	T	0.0735	ITGA8
rs10748818	A	-0.079	GBF1
rs72840788	A	0.0763	BAG3 (high LD with p.C151R)
rs117896735	A	0.4354	INPP5F
rs7938782	A	0.087	RNF141
rs12283611	A	-0.0645	DLG2
rs3802920	T	0.1073	IGSF9B
rs76904798	T	0.1439	LRRK2
rs34637584	A	2.4289	LRRK2 (LRRK2 G2019S)
rs7134559	T	-0.0539	SCAF11
rs10847864	T	0.1478	HIP1R
rs11610045	A	0.0601	FBRSL1
rs9568188	T	0.0617	CAB39L
rs4771268	T	0.0675	MBNL2
rs12147950	T	-0.0529	MIPOL1
rs11158026	T	-0.0842	GCH1
rs3742785	A	0.0707	RPS6KL1
rs979812	T	0.061	GALC
rs2251086	T	-0.1186	VPS13C
rs6497339	A	0.063	SYT17
rs2904880	C	-0.065	CD19 (CD19 p.L174V)
rs11150601	A	0.0907	SETD1A
rs6500328	A	0.0586	NOD2
rs3104783	A	0.0668	CASC16
rs10221156	A	-0.1156	CHD9
rs12600861	A	-0.0565	CHRNA1
rs12951632	T	0.0642	RETREG3 (high LD with MLX p.Q139R)
rs2269906	A	0.0631	UBTF
rs850738	A	-0.071	FAM171A2
rs62053943	T	-0.27	CRHR1
rs117615688	A	-0.2324	CRHR1
rs11658976	A	-0.0624	WNT3
rs61169879	T	0.082	BRIP1
rs666463	A	0.076	DNAH17
rs1941685	T	0.0531	ASXL3
rs12456492	A	-0.0983	RIT2
rs8087969	T	-0.0578	MEX3C
rs55818311	T	-0.0696	SPPL2B
rs77351827	T	0.0802	CRLS1 (high LD with CRLS1 p.P94L)

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rs2248244	A	0.0714	DYRK1A
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References

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2. Purcell S, Neale B, Todd-Brown K, et al. PLINK: A Tool Set for Whole-Genome Association and Population-Based Linkage Analyses. *Am J Hum Genet*. 2007;81:559–575.
3. Grenn FP, Kim JJ, Makarios MB, *et al*. The Parkinson's Disease: Genome-Wide Association Study Locus Browser. *Mov Disord* 2020; **35**: 2056–67.

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